

Research Proposal: Weighted Rolling Validation for Model Selection

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This research project studies the use of weighting in rolling validation (RV) for model selection. The direction of the project was developed with my sponsoring professor, and my work focuses on testing different weighting schemes, studying their theoretical behavior, and identifying conditions under which the weighted RV statistic agrees with the corresponding population prediction error.

The current problem considers two competing models: an oracle model, referred to as Model 0, and a mean model, referred to as Model 1. Although the population prediction errors of these two models can be compared directly, the model preferred by the rolling validation statistic does not always change at the same point as the model preferred by the population error. In other words, the crossing point of the RV criterion may not match the crossing point of the true population-level comparison.

The main goal of this project is to investigate whether an appropriate weighting scheme can make these two comparisons consistent. I study weighted versions of the rolling validation statistic and examine the expected difference between the two models,

$$\mathbb{E}[\text{RV}_{0,w}(n)] - \mathbb{E}[\text{RV}_{1,w}(n)].$$

When this quantity is negative, the weighted RV statistic prefers the oracle model in expectation. I then compare this result with the population prediction error difference,

$$R_0(n) - R_1(n),$$

where $R_0(n)$ and $R_1(n)$ are the population errors of Model 0 and Model 1.

The main question is whether there is a range of weights, or a general condition on the weighting sequence, such that the point where

$$\mathbb{E}[\text{RV}_{0,w}(n)] < \mathbb{E}[\text{RV}_{1,w}(n)]$$

agrees with the point where

$$R_0(n) < R_1(n).$$

Ideally, the crossing point from the expected weighted RV statistic should match, or become asymptotically equivalent to, the crossing point from the population prediction error. The purpose of introducing weights is therefore not simply to improve the numerical performance of RV, but to understand whether weighting can make the validation criterion reflect the same underlying model

comparison as the population error.

To study this question, I examine several classes of weighting functions and investigate how each one changes the expectation of the RV statistic. The project includes both theoretical calculations and simulation studies. The theoretical work is used to derive the expected weighted RV statistic and determine what properties the weights need to satisfy. The simulations are then used to check whether the theoretical behavior is visible in finite samples and whether the crossing points of the two comparisons become closer under the proposed weighting schemes.

The expected outcome of the project is a clearer characterization of the conditions under which the weighting works. Rather than identifying only one specific weight that performs well, the goal is to understand the general behavior of the weighting sequence and determine what type of growth or decay is needed for the weighted RV statistic to be consistent with the population-error comparison. This would provide a more useful theoretical result because the condition could potentially apply to a broader class of weighting functions.

One strength of the project is that it combines theory and simulation. The theoretical analysis helps explain why a certain class of weights should work, while the simulations provide a practical way to evaluate the behavior at realistic sample sizes. This also allows the project to distinguish between an asymptotic result and what can actually be observed in finite samples.

At the same time, the current research design has some limitations. The analysis focuses on a relatively simple comparison between an oracle model and a mean model. This setting is useful because the relevant expectations and population errors can be studied analytically, but the conclusions may not immediately extend to more complicated models. In addition, agreement in expectation does not guarantee that the correct model will be selected in every individual sample, since the realized RV statistic is still affected by random variation.

After the expectation-based result is established, a natural next step would be to study the variability of the weighted RV statistic and the probability of incorrect model selection. It would also be useful to examine whether the same weighting conditions continue to hold for more complicated candidate models, different data-generating processes, and other rolling-validation settings.

Overall, the purpose of this project is to determine when weighting can make the expected rolling-validation criterion agree with the true population-level comparison between competing models. By identifying the conditions that the weights must satisfy, the project aims to provide a clearer theoretical understanding of how weighted rolling validation can be used for more reliable model selection.